

# In/As/With 3D

## Developing a Framework for 3D Information Visualisation

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### Abstract

The notion of mapping information three-dimensionally has caught the attention of many researchers for the past few decades, especially in the field of computer science. However, many of their focuses are on exploring the potential of emerging technologies, and not on its relevance or how it can be different from conventional visualisation techniques. This paper argues that key principles of 3D information visualisation are not bound to any specific technology but lie in our physical and cultural conception of space and objects. To further propel this view, this paper proposes three approaches to designing a 3D information visualisation: *in/as/with* 3D. Through assessing and structuring discussions from various disciplines to explore possible strategies of each approach, this paper provides a foundational framework to discuss and develop this notion further.

### CCS Concepts

• **Human-centered computing** → **Visualization theory, concepts and paradigms**; *Information visualization*; *Virtual reality*.

### Keywords

3D information visualisation, tangible interaction, spatial conception, design principles

#### ACM Reference Format:

Akihiko Sugiura. 2025. In/As/With 3D: Developing a Framework for 3D Information Visualisation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3720053>

## 1 Introduction

### 1.1 Overview of prior studies

With the advances in technology and the distribution of virtual reality (VR) hardware over the past decade, many opportunities have opened up for designers to conceptualise its new standards. How can information be perceived differently in 3D virtual environment (VE) than with 2D flat screens? What unique information acquisition experience could VE present to its users? With the third

dimension being unlocked, there is a growing ambition among the community to present something beyond what paper can offer.

The use of the additional dimension in information visualisation is diverse across different domains. For example, the current VR market products tend to remain transferring 2D techniques to the 3D environment. The home screens for VR platforms are often composed of several windows floating perpendicular to the floor as if everything was projected on an invisible 2D plane. Most examples from existing VR programs fall into one of the four categories of displaying information in VEs proposed by Bowman et al. [2]: world-fixed, object-fixed, display-fixed, and user-fixed. Either way, the use of 2D windows remain the same: the information may be located *in* 3D space, but is not visualised *as* 3D structure.

With regard to visualising information *as* 3D structure, much research has been conducted in the field of computer science and human-computer interaction (HCI). Some research justifies the use of the third dimension as a way to display more information inside the screen space [17], while others see its potential to reduce path crossings in network visualisation [25]. Yet, as ambitious as their projects are, they do not necessarily meet the criteria of good information visualisation. Some of the problems are due to the limitation of displaying a 3D structure in a 2D screen, or are caused by lack of understanding of human factors (e.g., occlusion of components, absence of visual cues in the 3D structure, or lack of clarity in navigation).

The issues listed above indicate that interacting *with* information visualised *in/as* 3D may require a different strategy than its 2D counterpart. Here, much research emphasises the importance of movement: viewers' capability to turn and manipulate 3D objects and/or to walk through the 3D virtual environment provides numerous viewpoints that are crucial to understanding 3D structures [17, 25]. Nevertheless, it should be noted that 3D interaction does not necessarily equal to using 3D input devices [2]. For instance, in the field of video games, a 3D model of a character can be controlled either by 2D input and button devices (e.g., keyboards), replicas of existing devices (e.g., steering wheels), or 3D tracking of user's motion and gestures (e.g., Kinect) [12]. While they stand on the same premise that users tend to project their own body when interacting with virtual 3D objects/environment [20], they also highlight a potential gap between the medium and control.

### 1.2 Missing perspectives

While the notion of 3D information visualisation has been tackled from different angles, it is nevertheless difficult to compare these attempts and arguments as the term "3D" is not understood in the same manner. The examples from the previous section highlight

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720053>

different types of three-dimensionality being discussed; that of the content, that of the medium which the content is represented in, and that of the interaction with the chosen medium. It is confusing because these three can be either 2D or 3D independent from each other. For example, Robertson et al.'s [17] 3D cone tree comprises of a 3D content (the topology of data), a 2D medium (flat PC screen), and a 2D interaction (keyboard and mouse). The current discussion around 3D information visualisation does not distinguish what exactly is turned into 3D, making it difficult to grasp the overview of this notion.

The confusion of these types may root in the rather tech-driven attitude among researchers and practitioners. Many prior projects couple 3D information visualisation with digital software and hardware as indivisible, making it harder to discuss the concept apart from the characteristics and the limitations of its medium. Their focus has been mainly on exploring the emerging computer technology, and not necessarily on its viewers. This can also be problematic as it ignores critical questions such as “what does it mean to think *in/as* 3D?” or “why do we even need 3D information visualisation?”

### 1.3 Scope of this paper

Concerning the discussion above, this paper aims to collect discussions and key principles of 3D information visualisation across different domains and structure them into a coherent framework to elucidate the design opportunities and challenges of this notion.

I will delve into this topic from two directions. Firstly, I argue that 3D information visualisation is a topic on its own, and should be discussed independently from any specific technology. There is a foremost need to understand how people perceive three-dimensionality and information from a sense-making level. This paper questions whether 3D information visualisation is truly specific to the digital realm. If our perception of 3D structures is based on our embodied experience in the physical world, and considering how limited bodily interaction current digital medium could provide, it is beneficial to gather hints from non-digital field of studies and our real-life experiences.

Secondly, I will introduce three approaches to design a 3D information visualisation: *in* 3D, *as* 3D, and *with* 3D, following the distinction made in section 1.1. Regarding the different levels of three-dimensionality discussed before, the *in/as* 3D approaches tackle the three-dimensionality of the content, whereas the *with* 3D approach focus on that of the medium and interaction. See below for the revised definitions of these terms.

- ***In* 3D:** The spatial placement of the content of visualisation and how its position may influence the meaning it conveys
- ***As* 3D:** The three-dimensional topology of data and how that may improve viewers' sense-making
- ***With* 3D:** Viewer's interaction with the tangible, three-dimensional medium and how that may influence their experience of reading 3D information visualisation

In the following chapters, I will explore each approach and identify their key principles through referring to different domain knowledge (chapter 2), point out the limitations and potential future direction of this study (chapter 3), and finally present a short summary of this paper (chapter 4).

## 2 Framework

### 2.1 *In* 3D

In the field of (two-dimensional) graphic design, one can find many factual laws of viewers' reading patterns such as Gutenberg diagram, the Z-pattern or the F-pattern [13, 16]. These examples indicate how certain positions are loosely associated with meanings based on our physical attributes or cultural behaviours (e.g., title should be on the top left, page numbers ought to be located on the peripheral). In other words, the meaning is not only created by the sign itself, but also where it is placed.

In the same light, considering that 3D information visualisation is a method to position various items spatially to seek sense-making as a whole, the basic idea behind the *in* 3D approach is to find rationale behind why some items are better placed in certain spots. Through gathering numerous hints from mainly the field of human science, I propose three perceptual guidelines to discuss *in* 3D: *spatial hierarchy*, *visual hierarchy*, and *spatial conception*.

**2.1.1 *Spatial hierarchy.*** Certain spatial positions/axes/directions receive more attention than others due to our physical characteristics. For example, Drèze, Hoch & Purk [6], in the field of retailing and in-store shelf management, argue that when organising products on shelves, the amount of effort to move away from one's natural eye position correlated with the level of attention each shelf received. Or in a more abstract and conceptual level, Tuan [21], in the field of geography, collected cultural evidence of how the recognition of front/back is fundamental for humans as the position of human eyes make their frontal space primary visual and moving forward innate. These findings suggest that our physical attributes influence our predilection of spatial direction or position. In short, the visibility of a position is comparable to the importance it conveys.

**2.1.2 *Visual hierarchy.*** Besides spatial hierarchy, visual elements can solely navigate viewers' attention. When introducing the Gutenberg diagram in their book “Universal Principles of Design”, Lidwell, Holden & Butler [13] pointed out how this pattern is valid for heavy text, i.e., visually homogenous information, but can be easily overridden by other visual elements and properties (e.g., use of different fonts, colours, and images) and by the composition of the page. The same can be said for 3D information visualisation: if one item is visually dominant, or if the whole structure is distorted in a way that one part stands out from the rest, readers are likely to start reading from that point regardless of where it is located.

Based on related works from graphic design, comic art, and studies of Gestalt principles (e.g., [1, 7, 14, 19]), visual hierarchy can be designed in two different levels. Firstly, there is the properties of individual items, such as size, shape, and colour. These properties can be utilised to catch viewers' attention in contrast with other items. Secondly, there is the properties of the whole structure, such as proximity and distortion. The spatial pattern created by the positioning of items may also be applied for navigation.

**2.1.3 *Spatial conception.*** Apart from aspects of visual perception discussed above, spatial positions/axes/directions themselves can convey cultural meanings. Evidence to support this can be found in the use of orientational metaphors in various languages [11, 21]. For

example, “*top* priority” implies the association between high and the degree of importance; “putting *aside* our differences” connotes not being in the centre as being subsidiary; and “the reasons *behind* his decision” signifies the spatial relationship between the result and the reason.

As Lakoff & Johnson [11] argue, metaphors are not mere poetic expression but an attempt to conceptualise the unknown by relating them to simpler and more direct phenomena such as spatial positions. Conversely, these orientational metaphors can serve as guidelines for spatially positioning items in regard to the relation with others: e.g., comparable information should be placed side by side; the basis of arguments should be displayed on the base of other structures. 3D information visualisation has the potential to literally illustrate orientational metaphors, to visualise information akin to how we conceptualise it.

**2.1.4 The balance between guidelines.** Similar to the examples of reading patterns in graphic design, the three perceptual guidelines of the *in* 3D approach explained above are intended to resolve a reading pattern for 3D information visualisation through understanding our association of meaning and space.

The interrelation between these three guidelines is hard to define. Ideally, they would compensate for each other. To provide a simple example from our everyday life, imagine designing a section for discounted books in a bookstore. You would first gather the books in one place, prepare an “on sale” sign that is bigger than the books (*visual hierarchy*), hang the sign from the ceiling to indicate that the books below the sign are the discounted items (*spatial conception*), and finally adjust the sign to the eye level of customers in order to catch their attention (*spatial hierarchy*).

Nevertheless, these three may not necessarily harmonise depending on the context. For instance, when viewers are obliged to read from the ground and continue upwards to their eye level, i.e., when spatial hierarchy contradicts spatial conception, visual hierarchy can be utilised to strengthen either side. An everyday example of this case would be road markings. While they serve as the foundation of traffic control (*spatial conception*), they are placed far below the natural eye position (low *spatial hierarchy*) and, therefore, are counterbalanced by the use of high-contrast colours (*visual hierarchy*). The balance between these three depends on the content of the information, the context of usage, and, ultimately, on the designer’s decision.

## 2.2 As 3D

As 3D approach questions when and what kind of content is worth visualising as a 3D structure. To clarify, the three-dimensionality under discussion here is not something ostentatious as the so-called 3D effects (which often turn out as a hindrance for precise reading [26]), but rather about the 3D topology of information. This paper will also not look into purely numerical data with three variables that can be mapped spatially as a 3D plot, or data related to physical objects (e.g., plans of architecture) that inherently involve 3D. Instead, it will focus on mapping abstract (verbal) concepts, and explore how the additional dimension can enhance the existing 2D information visualisation techniques that we have become so accustomed to ground our mental model [18, 22].

It should be noted, however, that it is inappropriate to assess a 3D structure without considering how it is being represented to and interacted with its viewers, because its legitimacy is solely on the easiness of human perception. Nevertheless, postulation of meaningful 3D structure of information is still possible. In the following sections, I propose two types of strategies to turn 2D structure into 3D and improve viewers’ sense-making: *adding height to 2D*, and *aggregation of different 2D information visualisation*.

**2.2.1 Adding height to 2D.** Network, a method to depict related nodes by connecting them with links, can often become entangled and impenetrable as the number of nodes and links increase. As briefly explained in section 1.1, several prior works in HCI have considered adding height as a way to provide easier comprehension in such case. Adding height to a 2D plane can decrease the number of path crossings and reduce its visual complexity due to more flexibility in space [25], help its viewers grasp the whole structure at once [17] and highlight emergent patterns that otherwise would have been difficult to notice.

**2.2.2 Aggregation of 2D information visualisations.** In mathematics, any numerical data with three variables could be plotted three-dimensionally. However, the number of variables and dimensions do not necessarily correlate in information visualisation. Take Minard’s acclaimed portrayal of Napoleon’s army in the Russian campaign of 1812 [15] for example. Minard’s work juxtaposes multiple variables: the number of troops, location, temperature, all into a single 2D plane, and still successfully explain Napoleon’s failure coherently. Visual properties (e.g., colours and shapes) also play an important role to represent and differentiate multiple variables on 2D graphs and matrix.

Yet, regardless of how these techniques serve their purposes, there is a limit to how much information can be compressed into a single plane, and it is often the case that we gather several charts, diagrams, texts, etc. to form an interpretation or decision. For instance, a geography exam in high school may ask its examinees to make an inference out of different visualisations (e.g., *compare the map of city X and the age population charts of each of its districts. What can be hypothesised about the impact of city X’s public transportation network on its neighbourhood residents?*). Here, students are tested whether they can mentally synthesise the two sources correctly.

The second strategy of *as* 3D is an attempt to externalise this process of merging multiple 2D information visualisations into one while preserving their original structure as much as possible. It is a meta-visualisation of how different 2D visualisations are related to each other so as to avoid misinterpretation among viewers. Imagine 2D visualisations as card sheets and positioning them in parallel or orthogonally to create a 3D structure. The synthesis of different 2D visualisation can be made by either connecting mutual items between them, and/or by mapping their conceptual relationship according to *spatial conception* from the *in* 3D approach.

Examples of such strategy is scarce but can already be found in exploratory projects. For example, in their attempt to explain the temporal and geographical spread of SARS-CoV-2, Yamabe, Nakagawa & Wakita [27] created a 3D structure which consists of a global map placed perpendicular to a phylogenetic tree. The two diagrams were combined by placing each node of the phylogenetic

tree at the location of their first detection, emerging a spatial pattern that explains the outburst of the virus.

## 2.3 With 3D

In comparison to *in/as* 3D, which described the relevance of three-dimensionality of the content from two different standpoints, *with* 3D approach investigates how such content should be materialised, and what qualities need to be considered for its design. Note that as this research explicate inside the bounds of non-digital examples, I will focus on tangible, i.e., physically existent and touchable medium. As pointed out in the previous section, the relevance of a 3D structure is dependent on what kind of interaction is made possible for its comprehension. To discuss this topic further, it is first important to acknowledge the advantages of having a tangible medium and enabling physical interaction.

**2.3.1 Unifying control and content.** Imagine three different depictions of a 3D structure: a drawing printed on a paper, a virtual model displayed on a computer screen, and a tangible model. How are these depictions different in terms of viewer's experience? Inspired by research on Tangible User Interfaces (TUIs) [9], the difference between viewing a static image on a paper, rotating a 3D structure inside a 2D computer screen, and physically rotating a tangible model can be explained as the degree of directness the viewer is given in manipulating information. A sheet of paper does not give any control over the depicted structure, a virtual 3D model in a computer screen decouples content and control, and a tangible representation serves as both content and control simultaneously. Regarding 3D structure, in which views from multiple vantage points are crucial to its full comprehension, having direct physical control over the content can be a huge advantage as it is most intuitive and minimise viewer's mental effort. For this reason, this paper will discuss three-dimensionality of medium and interaction together.

**2.3.2 Tangible as a way to think with hands.** Because content and control are unified in a tangible medium, the experience of reading information visualisation shifts from a passive, instant action of comprehension to a procedure that requires its readers to actively interact with. The design of medium does not feed information on its own, but rather provides scaffolds for thoughts [3, 10] that users can build upon. In this light, 3D medium can be considered more as an interface or a cognitive tool rather than a mere representation of data. This is an interesting expansion of the concept, as it implies that the design of the medium and interaction may influence our way of thinking separately from the content.

To consider reading and thinking as indistinguishable is a perspective partly shared by constructivists in pedagogy, who claim that understanding requires one's active participation [24]. This can happen on many levels: research in cognitive science, namely embodied cognition, emphasise the importance of physical interaction with our surroundings and tools [4, 5]. For example, Hutchins [8] explains how bodily movements associated with the use of tools can substitute internal calculation by describing sailors' use of plotting tools for ship navigations. In short, we tend to think with our hand (or feet, or any other bodily movements), and such physical interaction with tools help us think differently [23].

Regarding this research, the arguments above suggest that the physical interaction with tangible medium can be more than just rotating it. For example, similar to how children learn various structures through playing with building blocks, involving viewers in the (re)construction of the 3D information visualisation may add another layer of cognition to reinforce the understanding of its structure.

**2.3.3 Strategies for interacting with tangible representation.** As a summation of this section, I suggest three strategies for interaction *with* 3D information visualisation using tangible medium. These interactions inversely define the prerequisite of the design of the medium and its possible context of use.

- **Providing multiple vantage points:** Allow viewers to walk around or rotate the structure in order to provide multiple vantage points. It is important that these actions are done deliberately and not autonomously. This type of interaction can be achieved not by the design of medium itself, but rather by how it is exhibited.
- **Reconstructing the depicted 3D structure:** Allow viewers to reconstruct the structure with their hands. Here, the design of tangible medium should be made so that it can be disassembled into segments and reconnected later. For successful reconstruction, it is equally important to provide clear instructions for the reconstruction process.
- **Enabling modification and construction of a new 3D structure:** Allow viewers to freely explore the relationship of different 2D visualisation and draw their own inference through mapping the items spatially. Tangible medium for this strategy should share similar characteristics with the second strategy. In addition, it should impose a restriction on the volume of space available to create a sense of positional relationship within (e.g., above/below, centre/peripheral) for utilising the *in* 3D approach.

## 3 Discussion

### 3.1 Utilising *in/as/with* 3D

This paper introduced three approaches to discuss and design 3D information visualisation: *in/as/with* 3D. See also figure 1 for a pictorial explanation of these approaches and their possible strategies. *In/as/with* 3D can be employed individually or together, depending on the content of the visualisation and its context of use. An ideal scenario of their combined use could be illustrated as follows. Once the 3D topology of information is defined through the *as* 3D approach, designers must decide, e.g., which side of the structure should face the top or bottom. This is where the *in* 3D approach comes in, as it is a question of human perception of space. After the content is mapped spatially, the *with* 3D approach can be referenced to decide the appropriate interaction strategy and the medium suitable for its intended audience. Designers may examine different types of three-dimensionality by going through *in/as/with* 3D respectively.

When designing a 3D information visualisation, it must also be noted that unlike its 2D counterpart, which can be scaled effortlessly to fit the projected screen, 3D information visualisation does not have much freedom in resizing. This is because the size of the

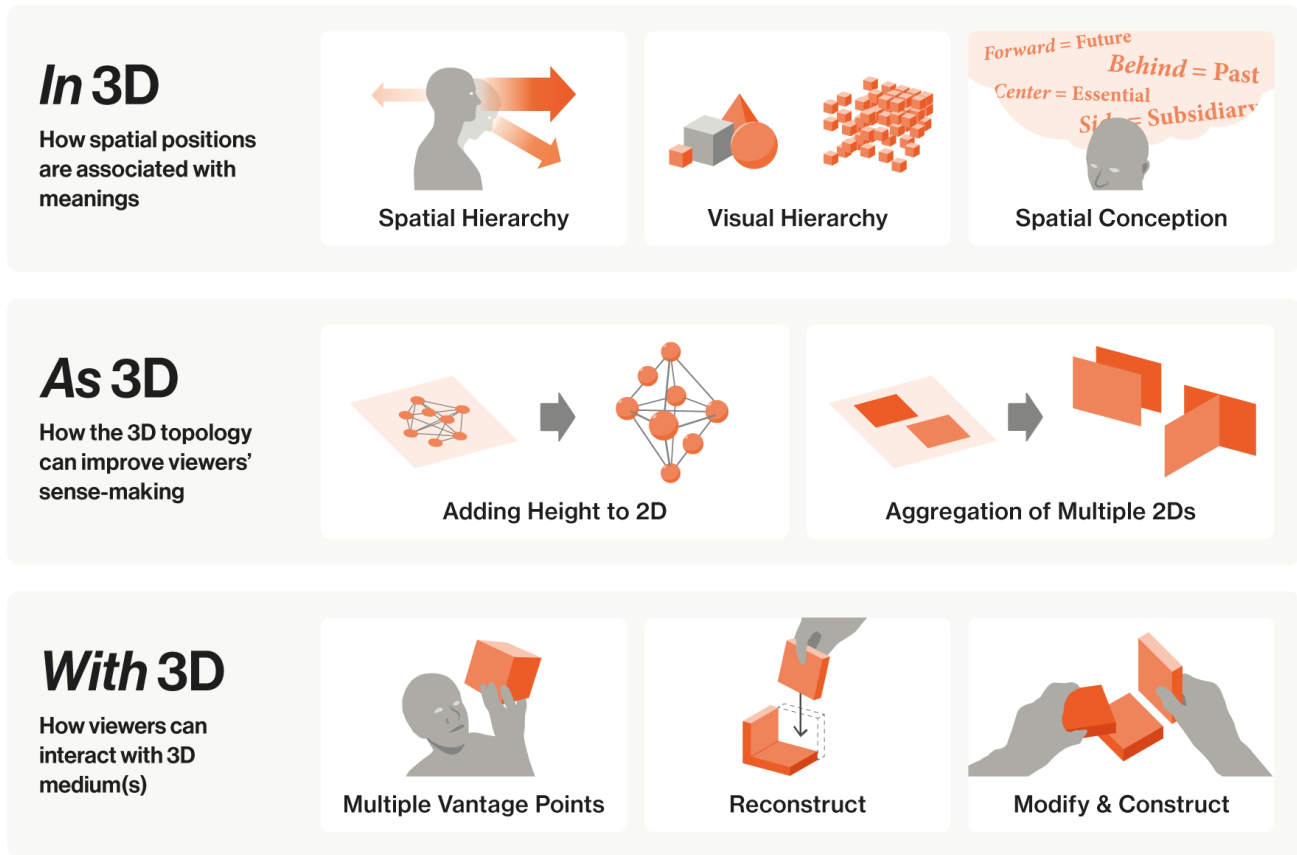


Figure 1: An illustrative summary of the *in/as/with* 3D and possible strategies for each approach.

tangible medium defines what kinds of physical interaction are made possible, i.e., viewers' reading and thinking experience can radically alter with scaling (see again, section 2.3.2). Considering that, *in/as/with* 3D can also be utilised to find the appropriate size of the visualisation. Questions such as "how can viewers best experience the relevant orientational metaphors?" (*in* 3D), "in which scale would the 3D topology of components be easiest to comprehend?" (*as* 3D), and "how many people are expected to view the visualisation at once?" (*with* 3D), can be helpful to decide the size and design a satisfactory experience for viewing 3D information visualisation.

### 3.2 Limitations and future directions

Although this paper took an interdisciplinary approach to gather evidence of 3D information visualisation from numerous perspectives, it did, however, overlook few relevant fields of research. Disciplines concerning spatial planning, namely architecture and interior design, were excluded as the focus of this paper was more on our conception of space itself. Yet, this excuse disregards our inclination of projecting our bodies to models of different scales. A similar idea already exists in mnemonic: the method of loci (or the memory palace) associates arbitrary information to a specific place for better

memorisation. Further research into these fields may provide rich insights that can be implemented into this discussion.

In addition, considering how cultural the conception of space is, the universality of some concepts presented here must also be questioned. While orientational metaphors can be found across various languages, the meaning assigned to a position can differ among them. Furthermore, as texts are often used to denote data, the variance of writing directions in language may influence users' affordance in interaction. Examining the cultural, linguistic aspect of 3D information visualisation will also be valuable to reinforce the validity of the developed framework.

### 4 Conclusion

While various attempts to visualise information three-dimensionally can be found across different domains, there still lacks a foundation to discuss this notion since the understanding of three-dimensionality is disparate. Concerning this issue, this paper propounded a coherent framework to structure and highlight key principles of designing 3D information visualisation. This paper began by distinguishing different types of three-dimensionality and argued that 3D information visualisation should be discussed independently from any specific technology. Furthermore, this paper

proposed three approaches to designing 3D information visualisation: *in/as/with* 3D, and investigated possible strategies for each approach.

Whereas technical discussions decay swiftly as new technologies are invented, human attributes remain elemental. The understanding of cultural conception of space and objects is therefore fundamental for designing a meaningful 3D information visualisation. This paper hopes to stimulate and empower other researchers and practitioners who also aspire to explore this notion.

## Acknowledgments

Part of this study was conducted at Köln International School of Design as part of the author's postgraduate research.

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